

Knowledge Base: Liquid Desiccant Spray Tower (LDST) System

Document Metadata

- **System:** Liquid Desiccant Spray Tower (LDST)
- **Desiccant Type:** Calcium Chloride (CaCl_2)
- **Cooling Capacity:** 6 TR (Ton of Refrigeration) / 21.10 kW
- **Primary Function:** Independent moisture removal from supply air using evaporative cooling, serving as an energy-saving alternative (up to 40% savings) to traditional vapor compression systems.
- **Applications:** Auditoriums, Textile Mills, Paper Industries.

CHUNK 1: System Architecture & Workflow

- **Dehumidification Process:** Process air comes into contact with a strong calcium chloride desiccant solution in the Conditioner. The desiccant absorbs moisture due to vapor pressure differences. This is an exothermic process, releasing the latent heat of condensation and chemical heat of mixing.
- **Regeneration Process:** The diluted desiccant is pumped to the Regenerator. It is pre-heated via a sensible heat exchanger. In the Regenerator, the liquid is exposed to regenerative air, transferring moisture from the solution back to the air. The now-concentrated solution is cooled and sent back to the Conditioner.
- **Key Circuit Components:** Conditioner Tower, Regenerator Tower, Evaporator Coil, Condenser Coil, Compressor, Expansion Valve, Heat Exchangers, Centrifugal Pumps, Blower Fans.

CHUNK 2: Structural & Geometric Specifications

Conditioner Dimensions:

- Height: **749 mm**
- Width: **249 mm**
- Length: **472 mm**
- Material: Polypropylene (Corrosion-proof, temperature-sustaining)

Regenerator Dimensions:

- Height: **722 mm**
- Width: **606 mm**
- Length: **322 mm**
- Material: Polypropylene (Corrosion-proof, temperature-sustaining)

CHUNK 3: Electromechanical Component Specifications

- **Conditioner Blower:** Airflow Capacity: 970 CFM (27.366 CMM / 0.4561 m³/s) | Function: Overcomes system static pressure to circulate conditioned air. | Mass of Air (ma): 0.4879 Kg/s
- **Regenerator Blower:** Airflow Capacity: 593.28 CFM (16.80 CMM / 0.280 m³/s) | Function: Supplies airflow to facilitate the thermal oxidation and moisture removal process in the regenerator beds.
- **Centrifugal Pumps (Solution Circulation):** Type: Chemical Slurry Pump | Maximum Flow Rate: 20 LPM (Operating: 18.189 LPM) | Maximum Head: 45 meters | Maximum Temperature: 120°C | Specific Gravity Handled: 1190 kg/m³ | Material: Polypropylene, PVDF with Mechanical Shaft Sealing.
- **Coils (Evaporator & Condenser):** Material: Stainless Steel tubing. | Evaporator Tubes: 33 tubes (Area: 1.923 m², Tube Diameter: 0.01905 m, Length: 1 m) | Condenser Tubes: 22 tubes (Area: 1.2879 m², Tube Diameter: 0.01905 m, Length: 1 m) | Heat Transfer Coefficient (U): 750 W/m²K
- **Storage Tanks:** Conditioner Tank Capacity: 224 Liters | Regenerator Tank Capacity: 300 Liters

CHUNK 4: Baseline Thermodynamic & Operating Parameters

(Note for AI Maintenance System: Deviations from these baselines indicate potential system degradation or failure).

System Design Calculations:

- Heat load (QH): **16.34 kW**
- Specific Heat of CaCl₂ (CpL): **2.0361 KJ/KgK**
- Specific Heat of Water (Cpw): **4.187 KJ/KgK**
- Specific Heat of Solution: **3.5417 KJ/KgK** (at 70% water, 30% LiCl assumption base)
- Solution Density: **1323.4 Kg/m³**
- Mass of Liquid Desiccant (mL): **0.4012 Kg/s**

Expected Temperature States:

- Process Air Input (Conditioner): **40°C**
- Process Air Output (Conditioner): **17°C** (at 50% Relative Humidity)
- Desiccant Regeneration Temperature Range: **80°C to 100°C**

CHUNK 5: Predictive Maintenance & Sensor Mapping Guide

- **Pumps (Flow & Vibration Sensors):** Centrifugal pumps must maintain a flow of roughly 18 LPM. Drops in flow rate, or increases in vibration, indicate potential impeller degradation, mechanical seal failure, or pipeline blockages.
- **Blowers (Pressure & Current Sensors):** The Conditioner Blower must maintain 970 CFM. Spikes in motor current (Amps) or drops in output static pressure indicate filter clogs, duct resistance, or motor winding wear.
- **Heat Exchangers (Temperature Sensors):** The system relies on precise temperature differentials across the evaporator and condenser coils. A narrowing temperature differential across the coils indicates scaling on the stainless steel tubes, frost accumulation, or refrigerant flow issues through the expansion valve.
- **Desiccant Tanks (Level & Concentration Sensors):** The 224 L and 300 L tanks must maintain adequate fluid levels. A rapid drop in concentration indicates poor regenerator performance (failure to reach 80°C - 100°C) or excessive moisture load beyond the 6 TR capacity.